

Strong baselines, controlled leakage, and a pre-specified inference gate for daily river water-temperature hindcasting

[AUTHOR LIST TO BE COMPLETED]¹

¹[AFFILIATION 1 TO BE COMPLETED — department or laboratory, institution, city, postcode,
country]

²[AFFILIATION 2 TO BE COMPLETED — add or delete affiliation lines to match the final author list]

Key points

- Daily river water temperature at 120 U.S. gauges is scored against damped persistence, a tree ensemble, and an LSTM on identical keys.
- The 2021–2023 evaluation labels are opened once, after the model suite, the inputs, and the analysis code are frozen and replayed.
- A pre-specified cluster gate fails on the frozen 15-region cohort, so all five formal comparisons are fixed-cohort descriptive effects.

Corresponding author: [CORRESPONDING AUTHOR TO BE COMPLETED],
[INSTITUTIONAL E-MAIL TO BE COMPLETED]

Abstract

Daily river water temperature is difficult to hindcast honestly. The signal is strongly persistent, so a copy of the most recent observation is already a hard reference; meteorological and hydrological covariates leak future information unless their issue-time availability is defined explicitly; sensor records are incomplete in ways that are correlated with season and with site; and gauges in the same region are not statistically independent. We assembled a panel of 657,480 site-days from 120 stable U.S. Geological Survey site numbers spanning 2006–2020, 34 states, and 15 two-digit hydrologic unit (HUC2) groups, and used it to evaluate ThermoRoute, a bounded-deviation predictor that combines a damped-persistence anchor, a learned relaxation proposal, a sparse horizon-conditioned variable and lag router, a strictly left-looking temporal convolutional encoder, a regime mixture, and a residual whose deviation from the anchor is algebraically bounded. Three design choices define the study. First, the comparison set is a strong baseline suite — persistence, damped persistence, seasonal climatology, an air2stream-style reference, a global gradient-boosted tree ensemble with and without site identity, and a global LSTM — scored on identical station/date/horizon keys, because a comparison against persistence alone is uninformative for this variable. Second, leakage is controlled mechanically rather than by assertion: all predictors are dated no later than the issue date, no horizon-specific future weather field is consumed, the temporal encoder is non-anticipating by construction, and every preprocessing, calibration, and threshold statistic is fitted strictly before the interval on which it is applied. Third, the 2021–2023 evaluation labels are downloaded exactly once, after the model suite and the input set are frozen, hash-sealed, and replayed by an isolated process that cannot reach the label namespace. A pre-specified, outcome-free inference gate requires at least 30 reportable clusters, an effective-cluster fraction of at least 0.75, and a largest-cluster share below 0.25. The frozen cohort contains at most 15 HUC2 groups with an inverse-Herfindahl effective cluster count of about 9.54, so the gate fails independently of any outcome. The five formal comparisons are consequently reported as fixed-cohort descriptive effects under the verdict `DESCRIPTIVE_ONLY_INFERENCE_GATE_FAILED`, with whole-HUC2 sign-flip p-values, cluster-bootstrap intervals, and Holm adjustments retained only as assumption-conditional sensitivities. We argue

that stating this limit explicitly, before the outcomes are seen, is more useful to the field than reporting cluster-naïve intervals as though they were inferential.

Plain Language Summary

River temperature changes slowly from one day to the next, so a forecast that simply repeats yesterday's reading is already fairly accurate. That makes it easy to publish a model that looks impressive but is not. We built a river-temperature model and then went to unusual lengths to test it fairly: we compared it against several strong alternatives on exactly the same days and sites, we made sure the model can never see information from after the moment it is asked to predict, and we agreed in advance — in writing, sealed against later editing — what would count as a result. We also decided in advance that our set of gauges is spread over too few large river regions to make statistical statements about rivers in general. That turned out to be true, so we report our numbers as a description of these 120 gauges rather than as a general conclusion. The final test data, covering 2021 to 2023, are downloaded only once, after everything else is locked.

Keywords: river water temperature; retrospective hindcasting; strong baselines; clustered inference; conformal prediction; pre-registration.

1 Introduction

Stream and river thermal regimes control dissolved-oxygen saturation, metabolic rates, life-stage timing, and habitat suitability, and they are set by a combination of atmospheric forcing, discharge, groundwater exchange, riparian shading, channel geometry, and regulation (Caissie, 2006). Daily water-temperature prediction has consequently become a common application for statistical and machine-learning models, from nonlinear air–water regressions (Mohseni et al., 1998) through hybrid air-temperature and discharge formulations (Toffolon and Piccolroaz, 2015; Piccolroaz et al., 2016) to deep sequence models, multi-task learners, and physics-guided river-network architectures (Feigl et al., 2021; Rahmani et al., 2021a; Jia et al., 2021; Sadler et al., 2022; Zwart et al., 2023). Recent reviews of this literature note that reported accuracies are difficult to compare because reference models, key sets, and data-splitting conventions differ across studies (Corona and Hogue, 2025).

The difficulty is not that daily water temperature is hard to predict. It is that it is easy to predict badly and still report an attractive number. Four properties of the problem make this so, and each one motivates a specific design decision in this study.

Persistence is a strong reference. Daily mean water temperature has a large thermal inertia relative to a one-day step. On the panel used here, the station-median root-mean-square error (RMSE) of naive persistence is 0.803 °C at a one-day lead and 2.217 °C at a seven-day lead, and simply relaxing the last observation toward a fitted seasonal climatology reduces the seven-day figure to 1.738 °C. Any model that is compared only against naive persistence therefore has roughly a quarter of the seven-day error budget handed to it before it learns anything. A reported skill score against persistence is not, by itself, evidence that a model has learned river thermodynamics; it may only be evidence that the model has learned to damp.

Issue-time leakage is easy and invisible. Water temperature covaries strongly with same-day and next-day air temperature. If a model is given a gridded meteorological field whose timestamp is the target date rather than the issue date, its apparent skill increases substantially, and nothing in a conventional train/test split will reveal this. The same problem arises more subtly through preprocessing: scaling constants, imputation fills, climatological references, event thresholds, and interval-calibration offsets are all statistics of the data, and fitting any of them on a period that includes the evaluated interval transfers information backwards in time.

Sensor records are incomplete, and the incompleteness is structured. In the frozen panel, roughly 15.8% of daily water-temperature values are absent, with maximum contiguous gaps of 2,404 days for water temperature and 1,369 days for discharge. Missingness of this kind is not missing-at-random: gaps cluster in winter at ice-affected sites and around instrument service intervals. Any evaluation conditioned on the observability of the target inherits that conditioning, and any station-selection rule based on coverage thresholds makes the resulting cohort an availability-enriched sample rather than a random one.

Rows from the same station and region are not independent. A daily panel of 120 gauges over fifteen years contains hundreds of thousands of rows, but it does not contain hundreds of thousands of independent observations. Treating daily

rows as exchangeable produces confidence intervals that are implausibly narrow. Treating stations as the sampling unit is better but still insufficient, because neighbouring stations share weather, geology, and regulation. The natural clustering unit for a U.S. panel is a large hydrologic region, and a study with a few dozen gauges typically has only a handful of such regions.

This study addresses the narrow empirical question that these four properties leave open: on fixed common station/date/horizon keys, at 1-, 3-, and 7-day leads, how does a constrained learned correction behave relative to a strong statistical reference set, to a tuned gradient-boosted tree ensemble, and to a global recurrent network, when the information boundary is enforced mechanically and when the evaluation labels are seen exactly once? It also asks whether the associated intervals and event probabilities are useful empirical descriptions of uncertainty. It does not attempt to infer an energy budget, a hydraulic state, or a management utility from the fitted architecture, and it does not claim prediction at sites that have no observational record.

Our contribution is therefore methodological as much as it is architectural. We report: (i) a comparison set that includes references most river-temperature studies omit; (ii) a leakage-control mechanism that is checkable rather than asserted; (iii) a regional-transfer design that holds out whole HUC2 regions and that we label honestly as *gauged* transfer, because issue-time water-temperature history remains available at the held-out sites; (iv) split-conformal intervals with an explicit statement of what they do and do not guarantee; (v) a single, irreversible acquisition of the 2021–2023 evaluation labels after the model suite is frozen; and (vi) a pre-specified inference gate on cohort cluster structure that we allowed to fail, and whose failure we report as a result about the study design rather than as a footnote.

The repository underlying this study also contains an older three-station case study using the identifiers b1, s2, and p3. b1, s2, and p3 are ordinary monitoring stations, not reservoirs. No verified metadata establish any upstream/downstream ordering, hydraulic connectivity, regulation status, or travel time among b1, s2, and p3. That legacy case is outside the present evaluation: its source provenance, station-identity metadata, measurement dictionary, and redistribution authorization are unverified, and its outputs predate the stable USGS site numbers and lineage

records used here. It is described in Section 2.5 only so that readers encountering it in the archive know its status.

2 Data and cohort

2.1 The frozen panel and its registry

The canonical development record is a single derived panel, `data_usgs/panel_usgs_120v2.parquet`, bound by a manifest (`data_usgs/frozen_panel_v1.json`) to a station registry (`data_usgs/station_registry_v1.csv`). Each of 120 sites carries one row for every calendar day from 2006-01-01 through 2020-12-31, giving 657,480 rows with no gaps in the date index; missing observations are explicit nulls rather than absent rows. Station identity is the zero-padded USGS `site_no`. Legacy sequential aliases of the form `n000–n119` exist in older artifacts and are never joined without first resolving them through the registry.

The registry spans 34 states and 15 HUC2 groups. The seven raw variables consumed by the models, in frozen order, are water temperature (`WTEMP`), stream-flow (`FLOW`), air temperature (`TEMP`), precipitation (`PRCP`), a relative-humidity proxy (`RHMEAN`), Daymet daylight-period mean incoming shortwave radiation in W m^{-2} (`DH`), and wind speed (`WDSP`). Two of these names are legacy and are easily misread: `DH` is Daymet `srad`, the incident shortwave flux averaged over the daylight period, not day length and not a 24-hour mean; `RHMEAN` is a reproducible vapour-pressure proxy evaluated at the `tmax/tmin` midpoint, not a direct daily-mean humidity observation. Gage height (`WLEVEL`) is retained as raw evidence but is not a model input. Water-temperature and discharge values come from USGS NWIS daily values (U.S. Geological Survey, 2024), meteorological fields from Daymet V4 (Thornton et al., 2022), and wind from gridMET (Abatzoglou, 2013).

Observed-value missingness in the committed panel is approximately 15.8% for `WTEMP`, 2.8% for `FLOW`, about 0.07% for `TEMP`, `PRCP`, `RHMEAN`, and `DH`, and zero for `WDSP`. These rates describe the derived panel; they do not reconstruct provider request history. The small meteorological gap is structured rather than random: all

120 sites lack the provider-calendar final day in the leap years 2008, 2012, 2016, and 2020. `WLEVEL`, retained only as provenance, is missing on 77.65% of rows.

2.2 Why the cohort is what it is

The cohort was not sampled at random from U.S. rivers, and the manuscript does not treat it as though it were. Candidate selection began from a discovery query over stream sites in the states represented by the panel and rejected 1,345 candidate stations: 950 lacked joint water-temperature and discharge availability, 376 failed a full-period coverage threshold, and 19 failed a 2019–2020 coverage threshold. The surviving ledgers are consistent with full-period coverage thresholds of 0.55 for `WTEMP` and 0.70 for `FLOW`, and 2019–2020 thresholds of 0.80 for both. This has an important consequence that we state plainly rather than bury: the 2019–2020 interval participated in cohort construction, so it is development data in the strict sense and has additionally informed model and narrative development. It is exploratory throughout this manuscript.

Two further properties of the retained cohort bear on the statistical treatment in Section 3.6. Among retained stations, 38 share a repeated HUC code with another retained station and 19 have another retained station within 10 km, so station-level independence is not tenable. Water temperature and discharge have maximum contiguous missing runs of 2,404 and 1,369 days respectively. Two stations contain 2,059 rows with signed negative discharge (minimum -121 cfs); these are retained with their sign rather than clipped, and their tidal, backwater, or measurement semantics are not resolved here.

The original discovery responses, retrieval timestamps, exact command line, and complete run configuration for the 1,465-candidate selection were not retained, so that step is auditable from committed ledgers but is not source-replayable. The panel also retains no NWIS qualifier columns, method or sensor history, or original daily-value response bytes, so measurement discontinuities and qualifier sensitivity cannot be reconstructed for 2006–2020. Both limitations apply only to the development record; the requirements for newly acquired evaluation inputs are stricter (Section 3.7).

2.3 Temporal roles

Temporal roles are fixed before any model is fitted, and a training sample is admitted only if its issue date and every one of its target dates fall inside the same partition.

Role	Dates	Rows	Observed WTEMP	Permitted use
Training	2006–2015	438,240	341,646	fit preprocessing and models
Validation	2016–2017	87,720	84,074	choose frozen settings
Calibration	2018	43,800	42,279	conformal and Platt fits; final year of the 2006–2018 seasonal event reference
Development evaluation	2019–2020	87,720	85,621	previously inspected; exploratory diagnosis only
One-time evaluation	2021–2023	—	sealed	opened once, after all freezes (Section 3.7)

On the development evaluation partition, the intersection of admissible keys across all primary models comprises 249,072 station/date/horizon keys, evenly distributed as 83,024 keys per lead. All development comparisons reported in Section 4 use exactly this common key set.

2.4 Evaluation-period inputs and the information boundary

The evaluation interval is 2021-01-01 through 2023-12-31 for the same 120-site cohort. Historical Daymet and gridMET covariates for that interval are retrieved and archived *before* any outcome is accessed. Meteorology is represented at each station coordinate and is not aggregated over upstream catchments, which is a real limitation for large basins and is listed as such in Section 6.

The primary information set uses provider values dated no later than each historical issue date and consumes no horizon-specific future weather forecast. This is a date-indexed retrospective hindcast. It does not replay as-issued conditions: the as-issued provisional vintage of a gridded product at a historical issue date cannot be reconstructed after the fact, so archiving requests, responses, timestamps,

and checksums freezes the dataset actually evaluated without proving that identical values were available operationally at that time. NWIS dates denote site-local finalized daily values while Daymet and gridMET use provider-specific calendar-day definitions; no subdaily day-boundary harmonization is claimed.

Because the 2006–2020 panel lacks its original HTTP responses, a separate outcome-free bridge re-fetched 2018–2020 Daymet and gridMET with the same parser used for the evaluation-period acquisition and compared every predictor on the exact site/date registry. Its committed manifest records `PASS_EXACT_PRODUCT_BRIDGE`. That gate detects product-version, parsing, scaling, missingness, and one-day alignment drift; it cannot prove that provider calendar days share NWIS local-day boundaries, and it cannot reconstruct as-issued availability.

A separate, deliberately restricted external cohort is defined after the model suite is committed. A fixed 34-state metadata query identifies stream sites advertising daily water-temperature capability, without requesting values, coverage dates, or event rates, and a deterministic seed selects 30 stations whose USGS site identifiers do not occur in the development registry. This is site-identifier disjointness only; it is not proof of HUC8, river-network, or distance separation. Models for that cohort are pooled and station-agnostic, but they still consume each target site's observed water-temperature history through the issue date. It is a history-dependent new-gauge exercise, it sits outside the formal comparison family, and it cannot change model or site selection.

2.5 Legacy three-station material

The archive contains three legacy CSV files identified as b1, s2, and p3. b1, s2, and p3 are ordinary monitoring stations, not reservoirs. No verified metadata establish any upstream/downstream ordering, hydraulic connectivity, regulation status, or travel time among b1, s2, and p3. Older outputs derived from those files predate stable USGS site numbers, the current panel contract, and the prediction-lineage records used here, and their source provenance, measurement dictionary, and redistribution authorization are unverified. Re-running the current pipeline would not bring that material inside the present evaluation. It is not evidence for any statement in this manuscript and it is not redistributed.

3 Methods

3.1 The predictor

ThermoRoute produces, for issue time t and lead h , a point prediction built from four parts. A damped-persistence anchor moves the last observed daily water temperature y_t toward a frozen seasonal climatology $c_{\{t+h\}}$ at a fitted rate and supplies the conservative reference trajectory. A learned flow- and season-conditioned relaxation proposal modifies that decay behaviour; its parameter is a fitted statistical quantity and is not interpreted as a measured residence time or heat-transfer coefficient. A sparse, horizon-conditioned router assigns weights over the seven variables at lags 0–14 following the sparsemax construction (Martins and Astudillo, 2016), and a strictly left-looking temporal convolutional encoder (Bai et al., 2018) with a regime mixture of experts (Shazeer et al., 2017) encodes recent history. Finally, the learned point correction is passed through $\tanh$ and scaled by a fixed `delta_scale`, so its deviation from the anchor is algebraically bounded.

Three clarifications matter for interpretation. First, "causal" in this paper describes time ordering only — the encoder is non-anticipating — and never causal inference. Second, the router is an allocation mechanism inside the predictor, not a map of connected reaches; it receives no verified graph or topology input and does not identify physical transport, travel time, residence time, or regulation. Third, the bound is relative to the named anchor: it yields a finite algebraic limit on the point output's deviation from damped persistence, and it bounds neither absolute error, nor event-tail error, nor interval width, nor behaviour after a distribution shift. The sequence builder supplies a 32-day tensor, but this is a construction buffer rather than an effective-memory claim: the current two-block, kernel-three encoder has a theoretical seven-step receptive field and the router's oldest usable value is lag 14, so no input older than lag 14 can affect the output.

The canonical configuration sets `delta_scale` to 1.0 °C. Because the 2019–2020 partition was inspected during development, this value is a development-selected algebraic point bound and is not presented as a selection from an untouched holdout.

3.2 A strong baseline suite

The most consequential methodological choice in this study is the composition of the comparison set. Much of the applied river-temperature literature reports skill relative to persistence, to climatology, or to an air–water regression. For a variable with this much day-to-day memory, those references are too weak to separate a model that has learned useful structure from a model that has learned to be conservative. We therefore fixed, before any evaluation, a suite in which each member removes a specific alternative explanation for apparent skill.

Persistence ($\hat{y}_{t+h} = y_t$) establishes the floor and quantifies the raw memory of the series. **Damped persistence**, fitted on the training period, relaxes y_t toward a frozen seasonal climatology and is the reference that matters: it absorbs the two effects — inertia and seasonality — that a learned model can most easily rediscover, and it is the reference against which the primary hypotheses are stated. **Seasonal climatology** isolates the seasonal-cycle component on its own. An **air2stream-style a4/a8 reference**, fitted by deterministic six-start bounded least squares on measured drivers, represents the hybrid air-temperature/discharge family that dominates the process literature; we describe below why its current status limits what can be said about it. **LightGBM** (Ke et al., 2017) is the principal learned reference and is run in two configurations: a global model that receives stable site identity as a categorical feature, and a per-station variant that isolates the value of pooling. A **global LSTM** with a station embedding represents the deep sequence family that has produced the strongest recent results for this variable (Rahmani et al., 2021a; Zwart et al., 2023).

The learned references are given a genuine chance to win. The global LSTM uses the same development splits, loss, station-balanced sampling, epoch and patience limits, history length, site identity, and five-seed budget as ThermoRoute, and its predeclared validation-only grid may expose the same frozen climatology, damped-anchor, and season features. It is eligible for the same conformal and probability calibration. It lacks only ThermoRoute's bounded-residual constraint and lag router. LightGBM selects among four predeclared candidate settings separately by lead on the 2016–2017 validation partition, using station-macro RMSE alone; the LSTM selects among three predeclared architectures with seed 0 before fitting five

members. All neural members — ThermoRoute, the global LSTM, and the plain-neural controls — are fitted by adaptive stochastic gradient descent with decoupled weight decay (Kingma and Ba, 2015; Loshchilov and Hutter, 2019) under a fixed schedule, and the optimizer, its schedule, and its stopping rule are identical across those model classes. These tuning budgets are documented and are *not* identical across model classes, which is a real asymmetry and is stated as a limitation rather than smoothed over. Consequently any statement about LightGBM in this paper is scoped to this frozen procedure — four candidates, this feature schema, this seed set — and is not a statement about gradient boosting in general or about a larger search.

The air2stream-style reference requires a specific caveat. The implementation used here is an unofficial style reference, not the official air2stream code and not a validated reproduction of it, and in the current artifact set it is recorded as `NOT_RUN`. A comparison that could speak to the published model would require the official implementation and its documented calibration search. We therefore report it as an unavailable, non-primary reference rather than quietly dropping it from the suite.

Beyond the primary suite, seven one-factor architecture controls (`DampedPriorOnly`, `TR-noDynamicPrior`, `TR-fixedKappa`, `TR-noRouter`, `TR-noMoE`, `TR-noTCN`, `TR-unbounded`) are fitted with the same five seeds as ThermoRoute and paired within seed on identical forecast keys and exact target values before ensemble averaging. A second control family (a plain multilayer perceptron and a plain causal temporal convolutional network) is information-matched as well as parameter-matched: both receive the same outcome-free issue-time history, station identity, climatology, frozen damped anchor, standardized environmental context, and calendar and regime-gate tensors available to the full model, and both predict unrestricted residuals around the same anchor while omitting the relaxation proposal, router, mixture, and residual bound. Neither ever reads the target tensor, the target-date field, or the disabled water-level channel. These are deletion and intervention sensitivities. They remove a direct information-set confound; they do not equalize historical hyperparameter-search budgets, prove component necessity, or identify a mechanism.

All comparisons use identical station/date/horizon keys for the two models being compared. No model-specific complete-case set is permitted, so a model cannot gain apparent accuracy by declining to predict on hard keys.

3.3 Leakage control

Leakage control in this study is a set of mechanisms, each of which can be checked against the code and the artifacts, rather than a statement of intent.

Issue-time separation of covariates. Every predictor consumed by a primary model carries a date no later than the issue date. No horizon-specific future weather field enters any primary model. An optional secondary analysis using archived GFS 2-m temperature composites is defined separately, is restricted by the archive's own start date, and is explicitly outside the primary registry.

Non-anticipating sequence encoding. The temporal convolutional encoder uses left-only padding, so output at position t is a function of positions $\leq t$ by construction rather than by convention. The router's admissible lag range is 0–14 and the encoder's receptive field is seven steps, so the effective information horizon is bounded and auditable, independent of the 32-day buffer used to assemble tensors.

Strictly backward-fitted statistics. Standardization constants, imputation fills, the seasonal climatology, the damped-persistence rate, the station-level q90 event thresholds (2006–2015), the seasonal event reference (2006–2018), the conformal offsets (2018), and the Platt calibrators (2018) are each fitted on data strictly preceding the interval on which they are applied. In the temporal arm these are station-specific and train-only; in the pooled arms they are pooled train-only transforms. This closes the most common backward-information channel in hydrological benchmarking, which is not the model but the preprocessing.

Explicit admissibility. A forecast key is admissible only when issue-date water temperature is genuinely observed and a 32-day history can be constructed. Other history cells may be filled with training-only seasonal medians and retain explicit missingness masks in the sequence input; no minimum observed-history fraction is imposed. We record one honest gap in this mechanism: the current issue-date auxiliary quantities used by the learned relaxation proposal and the regime gate

(standardized forcings, discharge, and the water-temperature tendency) are computed from the same training-only imputed panel but do not each carry a separate auxiliary-path mask. The model sees the corresponding masks through its sequence branch, but its output is not guaranteed invariant to the chosen fill values. This is a declared design limitation, not a claim that the auxiliary path is mask-isolated.

Transform conventions that do not encode outcomes. Discharge uses a signed `log1p`, preserving observed negative values rather than clipping them or relabelling them as drought; precipitation uses a nonnegative `log1p`.

Replay under isolation. Before any label acquisition is authorized, a fresh isolated interpreter running under `python -I -B`, with network access, child processes, repository writes, and reads from the evaluation namespaces all denied, reloads every trained member and every prediction head and reproduces the validation, calibration, and 2019–2020 development keys and values from the frozen inputs. The replay maps historical panel aliases through the registry, requires every selected station/target-date truth value to equal the frozen-panel value, independently recomputes the q90 thresholds and seasonal references, and refits the conformal and Platt parameters from the exact 2018 member-averaged rows. A prediction file that is merely self-consistent, or a re-hashed calibration object, is rejected. This is what prevents a rewritten artifact from substituting invented truth or calibration parameters for the real ones.

3.4 Regional transfer design

Random held-site splits are the usual way to report spatial generalization in this literature, and they leak. If a held-out gauge's neighbours remain in training, the model can reach the held-out site's thermal regime through correlated forcing, shared preprocessing statistics, and spatially smooth learned representations. We therefore report two arms with different leakage properties and label them differently.

The **random held-site warm-start** arm uses four folds in which held-site histories still contribute to global panel preprocessing. This is the arm most comparable to published random-split results, and precisely for that reason it is not the arm we treat as the spatial test.

The **held-region gauged transfer** arm packs the 15 HUC2/unknown groups into four folds of [30, 30, 31, 29] stations and holds out *whole regions*, so no gauge from a held-out region appears in training. Station-agnostic ThermoRoute and a global LightGBM are each trained on the in-fold regions and forecast the held-out region's stations. All climatology, scaling, and damped-rate parameters are pooled from in-fold stations only. The mean distance from a held-out station to its nearest training gauge is 289 km.

The label on this arm must be exact, because the distinction is routinely blurred in this literature. Held-out sites still provide their own observed water temperature through the issue date, both as the persistence anchor and as the sequence history. This is **gauged** transfer to a region whose gauges were not used in fitting; it is **not** prediction at a site that has no observational record, and no statement in this paper should be read as the latter. Such prediction would require the model to operate with no target-site water-temperature record at all, which is a different problem and a different evaluation (Weierbach et al., 2022; Rahmani et al., 2021b). The same caveat applies to the 30-site external cohort of Section 2.4, which is site-identifier disjoint but still history-dependent.

3.5 Conformal intervals

Predictive uncertainty in hydrological modelling has most often been represented through Bayesian or likelihood-based treatments of parameter and input error (Beven and Binley, 1992; Kavetski et al., 2006a, 2006b). We use a distribution-free calibration step instead, because the quantity required here is an empirical coverage statement about a fixed set of forecast keys rather than a posterior over model parameters.

Interval estimates are produced by split conformalized quantile regression (Romano et al., 2019; Vovk et al., 2005). The learned models emit a separate mean-squared-error point head and three pinball-trained quantile heads at levels 0.05, 0.50, and 0.95. Members are averaged with equal weights, and the conformal offset is fitted on the 2018 calibration year only.

The construction is deliberately one-sided in its effect. The exact split-conformal order statistic is retained as a signed raw audit value, but the deployed

offset is $\text{qhat_plus} = \max(\text{raw_qhat}, 0)$, and the delivered interval is $[\text{q05} - \text{qhat_plus}, \text{q95} + \text{qhat_plus}]$ with the median left unchanged. Calibration may therefore leave the nominal interval unchanged or widen it, but it can never shrink it. A non-finite offset, or a non-finite, crossed, or empty final interval, fails closed before any evaluation label is read. In the temporal arm offsets are fitted per station and lead; in the pooled arm they are fitted per lead only.

What this does and does not guarantee is worth stating precisely, because conformal prediction is frequently over-claimed in applied work. Split conformal provides finite-sample *marginal* coverage under exchangeability of the calibration and evaluation nonconformity scores. Here the calibration scores come from 2018 and the evaluation scores from a later, disjoint period at the same sites, so exchangeability is not satisfied by construction: there is temporal drift, and the scores are dependent within station and within region. We therefore report achieved coverage as an *empirical marginal diagnostic* and make no finite-sample guarantee. We make no conditional-coverage claim of any kind: nothing here establishes that coverage holds within a season, within a temperature regime, within a region, or at a particular station. Two pre-specified sensitivities probe that boundary — a block-maximum calibration variant that groups consecutive retained calibration rows, following the standard practice of resampling blocks rather than rows when the series is dependent (Künsch, 1989), and an idealized delayed adaptive-conformal variant that uses each forecast's target date as a feedback-arrival proxy. Neither replays real feedback availability, because the inputs contain no verified observation-publication timestamp, revision history, data vintage, or reporting latency; both report empirical marginal coverage only. The equal-weight three-quantile pinball summary is a three-quantile score and is not CRPS; the two are distinct members of the family of proper scoring rules (Gneiting and Raftery, 2007).

An event head reports exceedance of each station's 2006–2015 q90 water temperature, calibrated by one Platt map per lead fitted on 2018 only, against a seasonal event reference fitted on 2006–2018. That threshold is an absolute statistical tail diagnostic local to each station. It has no biological, ecological, or regulatory interpretation and is not comparable across stations. Both the Platt fit and the probability summaries give every retained station equal total weight, so stations with more retained days cannot dominate the calibration map.

3.6 Estimand, comparison family, and the pre-specified inference gate

The sampling unit is the station. For each lead, unweighted RMSE is computed on the common daily keys separately for each reportable station, and the primary effect is the unweighted median across stations of the paired difference $\text{RMSE}(\text{ThermoRoute}) - \text{RMSE}(\text{reference})$. A station/lead cell is reportable only with at least 100 valid paired targets. Daily rows therefore do not determine between-station weight, and a station with a long record does not dominate the estimand. This estimates performance conditional on observable issue and target water temperature within the frozen availability-enriched cohort. We report RMSE and paired RMSE differences rather than an efficiency-type criterion such as the Nash–Sutcliffe efficiency or its decomposition-based successors (Nash and Sutcliffe, 1970; Gupta et al., 2009), because those criteria normalise by a station-specific variance, which would make a paired between-model difference on identical keys harder to read.

The frozen comparison family contains exactly five rows:

#	Comparison	Lead	Predeclared margin
1	ThermoRoute vs. damped persistence	1 d	0.00 °C
2	ThermoRoute vs. damped persistence	3 d	0.00 °C
3	ThermoRoute vs. damped persistence	7 d	0.00 °C
4	ThermoRoute vs. LightGBM	3 d	+0.05 °C numerical ceiling
5	ThermoRoute vs. LightGBM	7 d	+0.05 °C numerical ceiling

The +0.05 °C ceiling was fixed before outcomes as a strict numerical limit on allowable degradation. It is not derived from sensor precision, biological response, a water-quality standard, or an elicited stakeholder utility, and it carries no ecological or regulatory importance.

Two uncertainty procedures accompany each row, both clustered at HUC2. Because the registry contains at most 15 clusters, the one-sided p-value is obtained by *exact enumeration* of all 2^K whole-cluster sign vectors rather than by Monte Carlo, applying one common sign to every station effect within a cluster. A 10,000-

draw whole-HUC2 cluster bootstrap gives the percentile interval for the median station effect, resampling complete clusters so that all station effects in a sampled region are retained together. Holm adjustment (Holm, 1979) covers exactly these five p-values. Exact enumeration removes Monte Carlo error but does not make the procedure distribution-free: it still assumes joint sign symmetry of each complete cluster effect vector around the tested margin. We do not use the standard tests of equal predictive accuracy for forecast series (Diebold and Mariano, 1995; Harvey et al., 1997), because their asymptotics are stated for a single loss-differential series and do not address dependence across the 120 gauges, which is the dominant dependence here.

This is where the study's central methodological decision was taken, and taken before any outcome was visible. An outcome-free amendment to the protocol overlays the original decision rule with a gate on the cluster structure of the cohort itself. The gate requires at least 30 reportable clusters, an effective-cluster fraction of at least 0.75, and a largest-cluster share below 0.25. Missing, unknown, or failed components fail closed.

The frozen cohort cannot pass it. Before any reportability attrition, the 15 HUC2 groups contain between 2 and 26 stations; the largest holds 21.7% of stations; and the inverse-Herfindahl effective cluster count is 9.54, an effective fraction of approximately 0.636. The cluster-count component fails outright and the effective-fraction component fails as well; only the largest-share component would pass on its own, and the gate is a conjunction. The verdict is therefore fixed by the geography of the frozen cohort, independently of the eventual numbers, and is permanently `DESCRIPTIVE_ONLY_INFERENCE_GATE_FAILED`.

The reporting consequences are strict and are followed throughout Section 4. Every one of the five rows is still rendered exactly once, including adverse, conflicting, or non-estimable outcomes; the effect estimate, the win rate, the station and cluster counts, and the interval are all shown. But the effect is a fixed-cohort descriptive quantity, and the exact sign-flip p-value, the cluster-bootstrap interval, and the Holm-adjusted value are retained solely as assumption-conditional sensitivities. They are not decision evidence. No row may be written as superiority, non-inferiority, equivalence, or parity, and no row may be generalized to a national

or U.S.-river population. Section 5.1 explains why we consider this a feature of the study rather than a defect.

Secondary and exploratory analyses — a qualifier-restricted target sensitivity, equal-HUC and leave-one-HUC influence summaries, station-balanced probability diagnostics, architecture controls, and synthetic missingness, noise, flow, and weather perturbations — are labelled as such and cannot promote or replace a formal row. Relative economic value in the cost–loss sense (Richardson, 2000) is recorded as `REV_NOT_EVALUATED_NO_PREDECLARED_COST_LOSS_RATIOS`: no cost–loss ratios, observed management costs, actions, or stakeholder utility model were frozen, so no economic value is computed, and a future cost–loss grid without those observed quantities would remain a hypothetical sensitivity.

A predeclared temporal coverage and calendar-balance audit accompanies each formal row and reports eight fixed descriptive candidates: equal weighting of the 12 year-by-season cells, leave-one-year for each evaluation year, and leave-one-season for DJF, MAM, JJA, and SON. The most adverse candidate-minus-reference effect is taken as the deterministic unfavorable value, with ties broken by frozen candidate order. The audit is restricted to keys where both issue-date and target-date water temperature are retained finite observations. It does not impute unavailable outcomes, test a missing-at-random assumption, estimate all-calendar-day performance, or establish stability across years or seasons, and a favourable sensitivity can never rescue an unfavourable primary row.

3.7 The one-time 2021–2023 evaluation

The single strongest methodological property of this study is that the evaluation labels are acquired exactly once, and that everything capable of being tuned is frozen and independently verified before that acquisition is permitted.

The evaluation interval is 2021-01-01 through 2023-12-31 for the frozen 120-site cohort. Those labels may not be read by model-selection, feature-selection, threshold-selection, calibration, or station-inclusion code. Enforcement is ordered, and the order is itself a gate. First, the protocol and its seal, the claim renderer and validator, the source tree, the dependency locks, and all temporal and station-agnostic model members are frozen and exactly replayed under the isolation de-

scribed in Section 3.3. Second, the model-suite registry is committed *while* the external candidate registry and the evaluation-period predictor artifacts are still absent from the repository. Only then may metadata-only candidate discovery and retrospective predictor acquisition run. Authorization replays the resulting Git ancestry and blob hashes and verifies that the model-suite commit genuinely precedes the candidate metadata and the evaluation-period predictor artifacts. Any source change after the model freeze invalidates authorization. Failure of the chronology gate does not produce a weaker result: it demotes the whole exercise to retrospective exploration and forbids the confirmatory reading entirely.

The acquisition itself is deliberately narrow and deliberately irreversible. A raw-only child process, using the USGS `dataretrieval` client for the National Water Information System (Hodson and Hariharan, 2023), is restricted to USGS daily-value requests for daily mean water temperature, discharge, and gage height. It records exact request and response bytes, series identifiers, approval qualifiers, final URLs, retrieval timestamps, and content hashes. Parameter and statistic identities are fixed in advance as daily mean (00003) water temperature (00010, °C), discharge (00060, cfs), and raw-only gage height (00065, ft). Every daily-mean series column is retained. On a given date, zero finite series values means missing; exactly one means that value, its qualifier, and its series identifier are used; two or more finite series means the parameter/date is marked missing with an explicit conflict code, with every constituent retained in the quality audit. Values are never averaged or selected across conflicting series, and **no site may be replaced after outcomes are accessed**.

The primary analysis uses every finite parsed daily-mean water-temperature and discharge value regardless of approval qualifier, so qualifiers can never select or remove a station, model, date, or forecast key. A separately labelled, non-confirmatory sensitivity — fixed before opening — retains target keys only when the target qualifier has the exact token set $\{A\}$, without refitting any model. Counts by cohort, station, variable, raw qualifier string, and value presence are reported immutably, and unknown qualifier strings are preserved and counted rather than interpreted.

There is one logical opening and one fixed request ledger. This is not a claim of exactly-once HTTP delivery, and we do not pretend otherwise: transport may retry, and a response received before its transaction directory is durable may be requested again. What is guaranteed is that a complete, durable, verifiable canonical response is never replaced; that a partial, invalid, or non-canonical transaction fails closed without overwrite; that cleanup is confined to unpublished owner-private state; and that once the acquisition manifest exists, raw network continuation is permanently disabled and only network-free deterministic recomputation may proceed. Normalized tables and the manifest are generated and validated in a private same-filesystem staging area and published by a single directory rename, as is the trusted scoring layer. These are honest-owner crash and replay guards; they are not protection against a malicious owner or a same-UID adversary, and we say so rather than implying a security property the design does not have.

The first successful scoring run writes a sealed result bound to the protocol commit, the data manifest hash, the resolved configuration hash, and the source-tree hash. A second scoring run after any model change is exploratory and must carry a new run identity.

3.8 Pre-registration and reproducibility

The analysis protocol was written and frozen before any post-2020 outcome was requested or inspected. It fixes the cohort and its hashes, the temporal roles, the model registry, the estimand, the five-row comparison family and its margins, the clustered p-value and interval procedures, the multiplicity correction, the probabilistic and event contracts, the outcome quality-control policy, and the decision rules — including the requirement that every row be reported exactly once regardless of direction. Two later amendments are themselves outcome-free and separately sealed: one narrows inferential wording through the cluster gate of Section 3.6, and one replaces the original single-seed rule for the architecture controls with the fixed five-seed matrix used here. A probability metric erratum, also sealed before any label access, corrects the source heads used for the three-quantile score. Protocol bytes are bound by SHA-256 to both their original preregistration commit and the final pre-label commit. We state the limit of that evidence plainly: the seals are repository-internal and assume an honest owner; there is no external timestamp,

public registration service, or independent custodian, and no local seal protects against an owner rewriting local history.

Reproducibility is receipt-based. Each training stage terminates in a content-bound completion record covering its members, heads, predictions, transforms, and lineage, and the model-suite freeze and the independent release verifier both require the complete set; a missing, stale, incomplete, or re-sealed partial record fails closed. Development reproduction is conditional on the committed panel bytes, because the original provider responses for 2006–2020 were not retained (Section 2.2). The full receipt chain, the fully transitive Python 3.12 dependency lock with package hashes, the environment probe, the protocol and amendments, and the per-stage manifests are archived with the software release (Section 8) and enumerated in the Supporting Information; the main text does not list individual receipts, stage numbers, or run identifiers, which belong in the archive rather than in the narrative.

4 Results

Section 4.1–4.5 report development-period quantities. **All of these are exploratory.** The 2019–2020 partition participated in cohort construction and has informed model and narrative development; it is not an independent test, and no number in these subsections is decision-eligible. Section 4.6 contains the formal rows, which are unfilled at the time of writing. Unless stated otherwise, all development values are five-seed ensemble means on the 249,072 common keys, with station-median RMSE in °C.

4.1 Point accuracy on the development partition (exploratory)

Lead	Persistence	Damped persistence	air2stream-style	LightGBM	LSTM	ThermoRoute
1 d	0.803	0.774	NOT_RUN	0.578	0.662	0.631
3 d	1.576	1.406	NOT_RUN	1.280	1.323	1.291
7 d	2.217	1.738	NOT_RUN	1.649	1.679	1.657

The shape of this table is the reason the baseline suite matters. Measured against naive persistence, the ThermoRoute ensemble's median station skill is +0.203, +0.187, and +0.251 at 1, 3, and 7 days. Measured against damped persistence, the same quantities are +0.168, +0.076, and +0.038. Roughly four-fifths of the apparent seven-day gain over persistence is attributable to damping alone. A study reporting only the first row of numbers would describe the same model very differently from one reporting both.

The tree ensemble has the lowest development-period station-median RMSE at all three leads. The paired station-level differences (`ThermoRoute` – `LightGBM`) are +0.046 °C at 1 day, +0.008 °C at 3 days, and –0.005 °C at 7 days, with ThermoRoute win rates of 0.00, 0.40, and 0.60 across the 120 stations. The 1-day win rate is reported as 0.00. We report this directly: on this partition, the constrained architecture does not have an accuracy advantage over a well-tuned tree ensemble, and its behaviour relative to that ensemble is lead-dependent. Against damped persistence, the paired median differences are –0.130, –0.104, and –0.062 °C with win rates of 0.86, 0.90, and 0.93.

The exploratory whole-HUC2 cluster-bootstrap intervals for those paired differences are [–0.177, –0.081], [–0.126, –0.083], and [–0.071, –0.052] °C against damped persistence, and [+0.043, +0.057], [+0.001, +0.020], and [–0.010, +0.010] °C against LightGBM. These are development-period mirrors of the frozen procedure. They carry the same 15-cluster structure that fails the gate of Section 3.6, and every row of the development mirror is flagged `NO_STRONG_INFERENCE` with the warnings `SMALL_CLUSTER_COUNT_LT_30` and `LOW_EFFECTIVE_CLUSTER_FRACTION`.

4.2 Regional structure and heterogeneity (exploratory)

Development skill against persistence is not carried by a single region or by a single size class. Region-weighted skill — the mean of the 15 per-HUC2 medians — is +0.204, +0.189, and +0.253 at 1, 3, and 7 days, essentially unchanged from the pooled medians of +0.203, +0.187, and +0.251. Per-region medians against persistence at 1 day range from +0.131 (HUC2:10) to +0.285 (HUC2:18); at 7 days from +0.217 (HUC2:06) to +0.290 (HUC2:09). Stratifying by drainage area into three groups of roughly 39 stations gives 1-day medians of +0.190, +0.214, and +0.231

from small to large. Against damped persistence, per-region 7-day medians compress to a range of +0.012 to +0.077, which is the same message as Section 4.1 restated spatially.

4.3 Transfer arms (exploratory)

Arm	Design	1 d	3 d	7 d
Temporal development	seen stations, 2019–2020	+0.203	+0.187	+0.251
Random held-site warm start	4 folds; held-site history in preprocessing	+0.178	+0.172	+0.241
Held-region gauged transfer	leave-HUC2-region-out	+0.155	+0.116	+0.155

Median station skill versus persistence. Against damped persistence the same three arms give +0.168 / +0.076 / +0.038 (temporal), +0.145 / +0.061 / +0.030 (random held-site), and +0.147 / +0.086 / +0.075 (held-region).

The ordering is the expected one and is informative in itself: skill degrades from the temporal arm to the random held-site arm and again to the held-region arm. The gap between the second and third rows is a direct measure of how much a random spatial split flatters a model on this panel.

In the held-region arm the global tree ensemble again attains lower station-median RMSE than ThermoRoute (0.652 vs 0.676 at 1 day, 1.391 vs 1.428 at 3 days, 1.786 vs 1.860 at 7 days), with paired median differences of +0.031 [+0.023, +0.040], +0.038 [+0.029, +0.050], and +0.067 [+0.049, +0.080] °C and ThermoRoute win rates near 0.16. The global LSTM, with its station embedding disabled in this arm, gives 0.679, 1.445, and 1.876. Every one of these arms retains issue-time water-temperature history at the held-out sites, so none of them speaks to prediction at a site that has no observational record.

4.4 Interval behaviour (exploratory)

Split-conformal intervals on the 249,072 development keys attain an empirical marginal coverage of 0.909 against a nominal 0.90, with a mean width of 3.87 °C and a mean interval score of 4.93. Per lead, coverage is 0.905, 0.910, and 0.912 with widths of 2.01, 4.22, and 5.38 °C. On the warm-season tail slice defined by training-period q90 exceedance (30,531 keys), coverage is 0.918 at a width of 3.50 °C.

Two pre-specified sensitivities show how sensitive these numbers are to the calibration convention. Calibrating on the maximum nonconformity score within blocks of seven consecutive retained calibration rows raises overall coverage to 0.981 but widens the interval to 5.74 °C — coverage bought with width, not with sharpness. The idealized delayed adaptive-conformal variants track nominal coverage more closely as the step size grows (0.903, 0.900, and 0.892 for $\gamma = 0.005, 0.02, 0.05$) but produce unbounded widths in several slices, so their interval scores are not finite. We report width and interval score beside every coverage figure precisely so that adaptive calibration is not presented as free. None of these figures is a conditional-coverage statement.

4.5 Component and input sensitivities (exploratory)

Deleting individual components changes development RMSE by small amounts: removing the temporal encoder is the largest single effect (0.631 $\rightarrow$ 0.679 at 1 day, 1.657 $\rightarrow$ 1.675 at 7 days), while removing the router, the mixture, the dynamic prior, or the residual bound moves the 1-day figure within ± 0.005 °C of the full model. Replacing everything by the damped anchor alone gives 0.774, 1.406, and 1.738. These are five-seed deletion sensitivities on paired keys; they do not establish component necessity or a capacity-matched attribution.

Synthetic corruption of issue-time inputs, applied without changing the forecast keys or the targets, orders the input channels by influence. Gaussian sensor noise at 0.25 and 0.5 training standard deviations degrades 1-day RMSE from 0.631 to 1.621 and 2.999 °C (+147% and +360%). Blocks of missing forcing values of 3, 7, and 14 days cost about +0.13 °C at 1 day and about +0.04 °C at 7 days. Air-temperature offsets of ± 2 training standard deviations cost +0.11 to +0.15 °C at

1 day. Multiplying discharge by 0.5 or 2 changes 1-day RMSE by at most $+0.003$ °C. The predictor is, on this panel, dominated by the water-temperature and air-temperature channels and is nearly insensitive to discharge perturbations. These are synthetic data-corruption probes; they are not climate projections, physically coherent scenarios, or deployment-safety tests.

Finally, the bounded-deviation contract was audited directly on the development partition. The pointwise anchor contract holds on 100.00% of rows, the derived pointwise error inequality holds on 100.00% of rows, the maximum absolute correction is 1.0000 °C against the configured 1 °C bound, and the derived station-by-lead RMSE inequality holds in all 360 station-by-lead cells. This verifies only that the implementation cannot move farther than the configured distance from its named anchor. It is not evidence of accuracy, extrapolation, calibrated uncertainty, or safety.

4.6 Evaluation-period results

The following slots are rendered deterministically from the verified acquisition receipt after the single opening described in Section 3.7. Each is reported exactly once, including adverse, conflicting, and non-estimable outcomes, and each carries the fixed-cohort descriptive verdict of Section 3.6. Handwritten substitution into these slots is rejected by body hashes and evidence bindings.

Table 4.1 — the five formal rows. Fields: model pair, lead, predeclared margin, reportable station count, reportable cluster count, median paired station RMSE difference, whole-HUC2 cluster-bootstrap 95% interval, win rate, exact sign-flip raw p-value, Holm-adjusted p-value, and the bound gate verdict.

#	Comparison	Lead	Margin	Effect and interval	Sensitivities (p _{raw} / p _{Holm})	Verdict
1	ThermoRoute vs. damped persistence	1 d	0.00 °C	[TO BE FILLED AFTER OPENING]	[TO BE FILLED AFTER OPENING]	DESCRIPTIVE_ ONLY_ INFERENCE_ GATE_FAILED
2	ThermoRoute vs. damped persistence	3 d	0.00 °C	[TO BE FILLED AFTER OPENING]	[TO BE FILLED AFTER OPENING]	DESCRIPTIVE_ ONLY_ INFERENCE_ GATE_FAILED
3	ThermoRoute vs. damped persistence	7 d	0.00 °C	[TO BE FILLED AFTER OPENING]	[TO BE FILLED AFTER OPENING]	DESCRIPTIVE_ ONLY_ INFERENCE_ GATE_FAILED
4	ThermoRoute vs. LightGBM	3 d	+0.05 °C	[TO BE FILLED AFTER OPENING]	[TO BE FILLED AFTER OPENING]	DESCRIPTIVE_ ONLY_ INFERENCE_ GATE_FAILED
5	ThermoRoute vs. LightGBM	7 d	+0.05 °C	[TO BE FILLED AFTER OPENING]	[TO BE FILLED AFTER OPENING]	DESCRIPTIVE_ ONLY_ INFERENCE_ GATE_FAILED

Table 4.2 — all-model scores on the evaluation common keys. [TO BE FILLED AFTER OPENING] — station-median RMSE for every primary model at every lead on the exact common key registry, with the reportable station count and the air2stream-style reference resolved as either fitted or unavailable.

Table 4.3 — temporal coverage and calendar balance. [TO BE FILLED AFTER OPENING] — for each of the five rows, all eight predeclared descriptive candidates (equal-weighted 12 year-by-season cells; leave-one-year for 2021, 2022, 2023;

leave-one-season for DJF, MAM, JJA, SON) and the deterministic most-adverse value.

Table 4.4 — interval behaviour on the evaluation keys. [TO BE FILLED AFTER OPENING] — achieved empirical marginal coverage, mean width, and interval score by model and lead, with the block-maximum and delayed adaptive-conformal sensitivities.

Table 4.5 — outcome quality control. [TO BE FILLED AFTER OPENING] — immutable counts by cohort, station, variable, raw qualifier string, and value presence; multiple-finite-series conflict counts; and the qualifier-restricted $\{A\}$ sensitivity for each formal row.

Table 4.6 — held-region and external arms on the evaluation keys. [TO BE FILLED AFTER OPENING] — leave-HUC2-region-out results and the 30-site site-identifier-disjoint arm, both labelled exploratory and both history-dependent.

Probabilistic metric suite — not reported. The pre-registered probabilistic metric stage does not execute on the frozen artifacts, for the reason given in Section 6.3, and its downstream analysis stage inherits the same condition. Interval evidence is reported from the split-conformal analysis instead.

5 Discussion

5.1 A pre-specified gate that failed is a result, not an excuse

The central inferential fact of this study is that we wrote down, before seeing any evaluation outcome, a condition under which our own headline comparisons would not be allowed to carry inferential weight — and that condition was not met.

The condition is about cohort geometry, not about performance. Clustered inference on a panel of gauges rests on resampling or sign-flipping whole clusters, and the accuracy of those procedures depends on having enough clusters, of reasonably comparable size, that the asymptotics they invoke are not a fiction. Fifteen HUC2 groups with an inverse-Herfindahl effective count of 9.54 and a largest group holding 21.7% of stations is not enough. This was determinable from the registry alone. It

required no outcome, and it could have been determined by anyone with the station list.

That is exactly why we consider stating it a contribution. The usual practice in applied water-temperature machine learning is to report a station-level or row-level confidence interval, or a paired test across sites, without addressing whether the number of independent spatial units can carry the procedure. Under that practice, a cohort like ours would produce intervals and p-values that look authoritative, and nothing in the paper would tell a reader that the effective number of clusters is under ten. The difference between that paper and this one is not the data and not the model; it is whether the limit is disclosed. We think the disclosed version is more useful, and we would rather publish a benchmark with an honest scope than an inference with a hidden denominator.

The practical consequence is that all five formal rows are fixed-cohort descriptive effects. The numbers are reported in full, because withholding them would be its own distortion, and the clustered p-values, bootstrap intervals, and Holm adjustments are reported alongside them as assumption-conditional sensitivities — quantities that describe what would follow *if* the cluster sign-symmetry and re-sampling assumptions held, which we do not assert. What they are not is decision evidence, and no favourable point estimate, narrow interval, or small p-value can convert them into decision evidence after the fact. The verdict was determined by the cohort before the outcomes existed, and it is not reopened by them.

There is a second instance of the same principle in this paper, described in Section 6.3, in which a pre-registered data contract rejected an output that we would have preferred to report. We left that contract alone as well. A pre-registration that never binds against its authors' convenience has not been tested; these two cases are the evidence that this one has.

The corresponding design lesson is concrete and transferable. If a study intends region-clustered inference over U.S. gauges, the cluster structure must be part of the *sampling design*, not a post-hoc grouping of whatever cohort availability produced. Reaching 30 or more effective clusters requires either a finer spatial partition with a defensible independence argument, or a deliberately stratified draw across many more regions, and either choice must be made — and its power computed — before

any outcome is accessible. The alternative route, of enlarging the cohort until the gate passes, was considered for this submission and set aside; retrofitting it onto the present frozen cohort would have meant choosing the cluster definition after seeing which definition helped, which is precisely the failure mode the gate exists to prevent.

5.2 What the development evidence does and does not indicate

Read as description of these 120 gauges over 2019–2020, the development results are informative in three ways and silent in a fourth.

They indicate, first, that the reference set determines the story. The same ensemble that shows median skill of $+0.251$ against persistence at 7 days shows $+0.038$ against damped persistence. Both numbers are correct. Only the second answers the question a reader of a modelling paper actually has.

Second, they indicate that a well-tuned gradient-boosted tree ensemble with site identity is a demanding reference on this panel: it attains the lowest station-median RMSE at all three leads in the temporal arm and in the held-region arm, with a reported ThermoRoute win rate of 0.00 at 1 day in the temporal arm. Reports in which a proposed architecture is compared only to persistence, climatology, or an air–water regression should be read with this in mind. The constrained architecture's practical argument here is not accuracy but the algebraic bound on its deviation from an interpretable anchor, which held in 100% of audited rows and in all 360 station-by-lead cells, and which is a property of the construction rather than a fitted outcome.

Third, they indicate that the spatial evaluation convention matters. Median skill against persistence drops from $+0.187$ to $+0.172$ to $+0.116$ at 3 days as the design moves from temporal, to random held-site, to whole-region holdout, with a mean nearest-training-gauge distance of 289 km in the last. A random spatial split on a panel where 19 stations have a neighbour within 10 km is measuring something closer to interpolation than to transfer.

They are silent on the question of prediction without target-site records. Every arm in this study, including the held-region and the site-identifier-disjoint arms, con-

sumes the target site's observed water temperature through the issue date, as anchor and as history. The problem of predicting at a location with no thermal record is a different problem, and this design cannot address it.

5.3 Relation to prior work

The comparison this study makes is narrow by construction. It is a common-key, clustered, pre-registered benchmark of a bounded correction against strong statistical and learned references at gauged sites. It is not an attempt to replace process-based thermal models, river-network graph models, or differentiable hybrid formulations (Jia et al., 2021; Rahmani et al., 2023; Zwart et al., 2023), nor to replace architectures that encode geographic context across regions and scales (Luo et al., 2025), each of which addresses structure — connectivity, upstream forcing, energy balance, spatial context — that a point-scale statistical predictor does not represent. Nor does the learned relaxation proposal recover any of that structure: it receives no verified graph or topology input and resolves no transport, travel time, residence time, or regulation.

Where we think the design is most portable is in the three controls that do not depend on the architecture at all: scoring every model on an identical key registry so that no model benefits from selective prediction; fitting every preprocessing and calibration statistic strictly backwards in time; and fixing the analysis, the margins, and the reporting rules before the evaluation labels exist. None of the three requires unusual computational resources. Together they change what a reported skill score means.

6 Limitations

6.1 Machine-verifiable scope statements

The following statements are part of the evidence contract and remain in the manuscript regardless of the eventual numerical outcome.

Route A is history-dependent and uses target-site water-temperature observations through each issue date; it does not establish ungauged prediction.

Route A does not establish independent river-network separation, physical river-network routing, or hydraulic travel time.

Route A is a one-shot retrospective historical-information evaluation, not an operational replay with archived as-issued predictor vintages or future NWP.

Route A uses daily-mean statistical thresholds and a numerical non-inferiority margin; neither has ecological, biological, or regulatory meaning.

Route A provides no physical, deployment, regulatory, or distribution-free safety guarantee, and failure to reject is not equivalence.

Route-A architecture controls and predictor sensitivities are descriptive or exploratory and do not identify causal mechanisms.

Route A evaluates a fixed availability-enriched 120-site cohort and is not nationally representative of all U.S. rivers or all calendar days.

Route A does not establish conditional coverage, and its equal-weight three-quantile pinball summary is not CRPS.

6.2 Cohort, measurement, and design limitations

The station sample is availability-enriched rather than randomly drawn, and the coverage thresholds that produced it used the 2019–2020 interval, so that interval is not independent of cohort construction. The original provider bytes for the 2006–2020 panel are unavailable, so development reproduction begins from the committed derived artifact and the 1,465-candidate discovery execution is auditable but not replayable. The panel retains no NWIS qualifier columns, method or sensor history for 2006–2020, so measurement discontinuities cannot be reconstructed for the development record. Meteorology is represented at the station coordinate rather than integrated over the contributing catchment, which is least appropriate for the largest basins in the cohort. Two stations contain 2,059 signed negative discharge rows whose semantics are unresolved. Outcomes are daily means, and the evaluation is conditioned on outcome observability. The auxiliary prior and gate features are computed from a training-only imputed panel without separate path-specific validity flags, so output invariance to fill values is not guaranteed. HUC2 is a coarse administrative grouping and is not an independent river-network component. The +0.05

°C margin is a numerical ceiling with no stakeholder-derived importance. Covariates are retrospectively acquired latest-provider values rather than as-issued vintages. Model-selection budgets were documented but not equalized across model classes, so any statement about a reference model is scoped to its frozen procedure. No latency, memory, energy, or multi-hardware benchmark is reported. The evidence chronology is owner-controlled, with no external timestamp or independent custodian. The air2stream-style reference has not been fitted and no official air2stream calibration run exists.

6.3 The probabilistic metric suite is not reported

The pre-registered probabilistic contract treats a zero-width nominal prediction interval as a fatal condition and prohibits evaluation-time repair. In the development panel, 135 of 26,993,675 member-level rows carrying complete quantile heads — 0.0005%, or about one row in 200,000 — have identical 5th, 50th, and 95th percentile predictions. All 135 come from the tree ensembles (123 from the global model, 12 from the per-station variant); no other model is affected. We emphasise what the artifact is *not*: there is no strict quantile-ordering violation anywhere in the panel, and the maximum monotonicity violation is exactly 0.000 °C. The rejected rows are correctly ordered but degenerate.

Their distribution is interpretable, and we flag the reading that follows as interpretation rather than as a measured attribution. The 135 rows fall on 12 distinct sites, all in hydrologic regions 04, 05, and 06 — the Great Lakes, upper Mississippi, and Missouri basins — and 129 of them occur at the 1-day lead, 6 at 3 days, and none at 7 days. In 95.6% of them the observed target is below 1 °C. This is the signature of ice-affected winter conditions, in which daily mean water temperature is pinned near the freezing point and the conditional 5th, 50th, and 95th percentiles genuinely coincide: for example site 04027000 at a 1-day lead, with all three quantiles equal to 0.0010 °C and to −0.0191 °C on different dates. A zero-width interval is arguably the correct answer under those conditions rather than a broken one. The 12 per-station cases, such as site 01542500 with all three quantiles equal to 13.1107 °C, are constant-leaf predictions from a model fitted on a small station-specific sample, which is an expected property of per-station gradient boosting. The model outputs are defensible; it is the contract's inclusive equality test that refuses them.

We did not amend the contract. It was frozen before any access to the evaluation labels, and relaxing it would have invalidated the frozen training receipts on which the chronology gate of Section 3.7 depends. The pre-registered probabilistic metric suite — interval coverage from the nominal heads, three-quantile pinball mean, reliability, and Brier score — is therefore not reported, and the analysis stage that consumes it is likewise not reported. Two things follow. First, interval validity in this paper rests entirely on the split-conformal analysis of Sections 3.5 and 4.4, which is unaffected: the conformal offset is non-negative by construction, so the delivered interval for every one of those 135 rows is strictly wider than the degenerate nominal one, and the deployed contract separately requires positive width. No delivered interval is degenerate. Second, the standard event-probability verification metrics — Brier score, reliability, discrimination, and calibration slope and intercept (Wilks, 2011) — are absent from this manuscript, and no claim about probabilistic calibration beyond empirical marginal interval coverage should be read into it. Resolving this requires a sealed erratum and a new training lineage, which is future work rather than a repair applied after seeing which resolution was convenient.

6.4 Scope of the observed-threshold analysis

An optional threshold analysis is separate from prediction evaluation. It can only compare independently sourced observed daily maxima, expressed as a strict seven-consecutive-day average of daily maxima, against a complete site-specific standards registry, with seasonal applicability assigned by each window's ending date. Season-external or incomplete windows remain unclassified, and missing, unrelated, or ambiguous standard matches fail closed. Any reported exceedance is descriptive: it is an observation, not a model result and not a legal determination, and it does not establish compliance with any law or regulation.

6.5 Archive scope

The active release namespace excludes withdrawn generated outputs and rendered manuscripts, but the self-contained history bundle deliberately retains reachable deleted objects so that chronology can be audited. It is not a byte-level purge. Those historical objects are provenance rather than current evidence and require a separate licence and privacy review before public redistribution. The current archive

profile is machine-marked local-owner evidence: public redistribution and third-party transfer are disabled and the public build mode fails closed pending a byte-bound rights review. This profile must not be read as a public software or data release.

7 Conclusions

We have described a river water-temperature hindcasting benchmark whose design choices are aimed at a single problem: making a reported skill number mean what a reader assumes it means. The comparison set includes damped persistence, a tuned gradient-boosted tree ensemble with and without site identity, and a global LSTM, all scored on identical station/date/horizon keys, because for a variable this persistent a weaker reference set does not discriminate. The information boundary is enforced by construction — issue-time-dated covariates, a non-anticipating encoder with a bounded lag range, and every preprocessing and calibration statistic fitted strictly backwards in time — and verified by an isolated replay that cannot reach the evaluation labels. Spatial generalization is reported from whole-region holdout and labelled as gauged transfer, because issue-time water temperature remains available at held-out sites. Intervals are split-conformal and are reported as empirical marginal diagnostics with width and interval score attached, with no conditional-coverage claim.

On the development partition, and as description of these 120 gauges only, the ensemble's median station skill against damped persistence is +0.168, +0.076, and +0.038 at 1, 3, and 7 days, against +0.203, +0.187, and +0.251 versus naive persistence; the tree ensemble attains the lowest station-median RMSE at all three leads; and skill degrades monotonically from the temporal arm through the random held-site arm to the whole-region holdout. The evaluation-period rows remain unfilled until the single acquisition described in Section 3.7 occurs.

Irrespective of those numbers, the formal comparisons will be fixed-cohort descriptive effects. The pre-specified inference gate on cohort cluster structure fails on a cohort with at most 15 HUC2 groups and an effective cluster count near 9.54, and that determination was made from the registry before any outcome existed. We report it as a finding about the design space of this literature: a benchmark of this

size cannot carry region-clustered inference about rivers in general, and reaching that regime requires the cluster structure to be part of the sampling design rather than a grouping applied afterwards.

Open Research Statement

Data availability. The derived daily panel (`panel.usgs.120v2.parquet`), the stable station registry (`station_registry_v1.csv`), the hydrologic-unit metadata snapshot, the candidate-rejection ledger, and the frozen panel manifest that binds them are deposited as one versioned dataset at [DATA DOI TO BE MINTED] under [DATA LICENCE TO BE ASSIGNED]. The evaluation-period acquisition record — exact request and response bytes, series identifiers, approval qualifiers, final URLs, retrieval timestamps, and content hashes — is added to that deposit as a new version after the acquisition described in Section 3.7.

Neither the DOI nor the data licence can be assigned before the byte-level rights review described in Section 6.5 completes. The derived panel encodes values from three providers whose terms differ, and a derived product does not inherit the most permissive of its upstream terms. Until every object proposed for the deposit carries a recorded, evidence-backed redistribution decision, the project's release tooling refuses to build a public archive. The deposit is therefore described here as planned and specified, not as existing.

Primary observational sources. All observations are obtained from public providers and none is the property of the authors. Daily-value water temperature and discharge come from the U.S. Geological Survey National Water Information System (U.S. Geological Survey, 2024; parameter codes 00010, 00060, and 00065 with statistic code 00003). Air temperature, precipitation, the relative-humidity proxy, and daylight-period mean incoming shortwave radiation come from Daymet V4 R1 at ORNL DAAC (Thornton et al., 2022). Wind speed comes from gridMET (Abatzoglou, 2013). Each retains its provider's own terms, citation requirement, and access route independently of this manuscript.

Material that cannot currently be redistributed. For any provider object whose redistribution terms are unresolved at deposit time, the deposit carries, in place of the bytes: the product identifier and version, the exact request specification (site list, parameter and statistic codes, and date range), the retrieval code, and a SHA-256 manifest of the bytes as retrieved. A third party can therefore re-acquire identical inputs from the provider and verify them against the manifest without relying on redistribution. At the time of writing the following classes have no recorded redistribution decision and default to exclusion: USGS NWIS response bytes; Daymet V4 subsets and any derived field that materially encodes Daymet values; gridMET responses and derived fields; and the mixed derived panel and registries built from them.

Two further classes are excluded outright rather than pending. The AGU LaTeX class used to typeset this manuscript is a third-party file supplied for submission only and is not redistributed in the archive. The three legacy CSV files described in Section 2.5 have no recorded source, collection terms, or redistribution authorization, and are not redistributed in any form.

Original provider responses and retrieval timestamps for the 2006–2020 development panel were not retained and cannot be reconstructed, so development reproduction begins from the committed derived artifact. That limitation does not apply to the evaluation-period acquisition, for which exact bytes are archived.

Software availability. The analysis software, the frozen protocol and its sealed amendments, the probability-metric erratum, the model-suite registry, the completion receipts, the per-stage manifests, the environment probe, the verification entrypoints, the deterministic result renderer that generates Section 4.6 from the acquisition receipt, and the fully transitive Python 3.12 dependency lock with package hashes are archived at [SOFTWARE DOI TO BE MINTED], corresponding to release [RELEASE TAG TO BE ASSIGNED] of the source repository at [REPOSITORY URL TO BE CONFIRMED].

The project's own source code is released under the MIT licence (SPDX identifier MIT). That licence covers the source code only. It does not license the observational data, the archived provider responses, the third-party typesetting class, or any redistributed dependency binary, each of which retains its own terms; third-party

components inside the archive are enumerated with their terms in a notice file, and any component whose terms are unresolved is excluded from the deposit rather than shipped under an assumed licence.

Analyses were run on Python 3.12 with NumPy, pandas, pyarrow, SciPy (Virtanen et al., 2020), scikit-learn (Pedregosa et al., 2011), LightGBM (Ke et al., 2017), PyTorch (Paszke et al., 2019), and the USGS `dataretrieval` client (Hodson and Hariharan, 2023).

Reproduction. Sections 4.1–4.5 are reproducible from the archived panel, the archived model bundles, and the pinned environment. Section 4.6 is reproducible from the archived acquisition record and the same bundles. Bit-level equality is expected for the tree models and agreement to a documented tolerance for the neural members; the tolerances, the verification commands, and the expected digests are listed in the Supporting Information. Reproduction has not yet been carried out by an operator independent of the authors on an independent host, and this statement is not a claim that it has.

Placeholder inventory. Five bracketed identifiers appear above: [DATA DOI TO BE MINTED], [DATA LICENCE TO BE ASSIGNED], [SOFTWARE DOI TO BE MINTED], [RELEASE TAG TO BE ASSIGNED], and [REPOSITORY URL TO BE CONFIRMED]. None is a real identifier, and none may be replaced by a reserved, draft, or example DOI. Together with the author block, the corresponding-author details, and the Acknowledgments below, they are the complete set of externally supplied fields that must be closed before submission; each is tracked, with the external input required to close it, in `docs/WRR.SUBMISSION_CHECKLIST.md`.

Acknowledgments

[FUNDING TO BE COMPLETED — each funder with its award number, or an explicit statement that the work received no external funding.]

[COMPUTATIONAL RESOURCES TO BE COMPLETED — the facility or facilities on which the model suite was fitted.]

We acknowledge the U.S. Geological Survey for the National Water Information System, the ORNL Distributed Active Archive Center for Daymet V4 R1,

and the Northwest Knowledge Network for gridMET. Acknowledgment of a data provider is not an endorsement of this analysis by that provider.

Competing interests. [COMPETING INTERESTS TO BE COMPLETED — a declaration is required from every author, including the explicit statement that none exists if that is the case.]

Author contributions. [CREDIT ROLES TO BE COMPLETED — assign each author to the applicable CRediT roles: conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, writing — original draft, writing — review and editing, visualization, supervision, project administration, funding acquisition. The signed intake form is `docs/FAIR_SUBMISSION_READINESS_AND_TEMPLATES.md` §2.]

Supporting Information

Supporting Information accompanies this manuscript and contains: the cohort and registry description with the candidate-rejection ledger (SI01); the issue-time information boundary and the product-compatibility bridge (SI02); model equations and unit conventions (SI03); the sealed protocol, its amendments, and the probability-metric erratum (SI04); the registered five-row family and its fields (SI05–SI06); all-model scores on the exact common keys (SI07); probability and reliability schemas together with the non-reporting disposition of Section 6.3 (SI08); architecture and information-matched controls with seed and budget provenance (SI09); the temporal coverage audit (SI10); spatial and leave-cluster sensitivities (SI11); outcome quality control and qualifier evidence (SI12); the history-dependent external arm (SI13); missingness and failure cases (SI14); reproduction hashes, commands, and environment parity fields (SI15); and the rights and data dictionary (SI16).

Nine supporting figures accompany these sections. Figures S1–S3 describe the frozen cohort and registry geometry, the temporal roles and issue-time information boundary, and the model and calibration dataflow; they contain no evaluation-period quantity. Figures S4–S8 expand the evaluation-period results of Section 4.6 — point-performance heterogeneity, probability diagnostics, temporal opportunity and attrition, spatial and leave-region influence, and outcome quality control with the external arm — and are rendered only after the acquisition of Section 3.7.

Figure S9 is an optional development-period conformal-calibration sensitivity, is labelled as such in the figure itself, and must not be compared numerically with any evaluation-period figure.

References

- Abatzoglou, J. T. (2013). Development of gridded surface meteorological data for ecological applications and modelling. *International Journal of Climatology*, 33, 121–131. (gridMET / METDATA) doi: 10.1002/joc.3413
- Bai, S., Kolter, J. Z., & Koltun, V. (2018). *An empirical evaluation of generic convolutional and recurrent networks for sequence modeling*. (arXiv:1803.01271)
- Beven, K., & Binley, A. (1992). The future of distributed models: model calibration and uncertainty prediction. *Hydrological Processes*, 6(3), 279–298. doi: 10.1002/hyp.3360060305
- Caissie, D. (2006). The thermal regime of rivers: a review. *Freshwater Biology*, 51, 1389–1406. doi: 10.1111/j.1365-2427.2006.01597.x
- Corona, C. R., & Hogue, T. S. (2025). Machine learning in stream and river water temperature modeling: a review and metrics for evaluation. *Hydrology and Earth System Sciences*, 29, 2521–2549. doi: 10.5194/hess-29-2521-2025
- Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13, 253–263. doi: 10.1080/07350015.1995.10524599
- Feigl, M., Lebiedzinski, K., Herrnegger, M., & Schulz, K. (2021). Machine-learning methods for stream water temperature prediction. *Hydrology and Earth System Sciences*, 25, 2951–2977. doi: 10.5194/hess-25-2951-2021
- Gneiting, T., & Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102, 359–378. doi: 10.1198/016214506000001437
- Gupta, H. V., Kling, H., Yilmaz, K. K., & Martinez, G. F. (2009). Decomposition of the mean squared error and NSE performance criteria: implications for improving hydrological modelling. *Journal of Hydrology*, 377, 80–91. doi: 10.1016/j.jhydrol.2009.08.003
- Harvey, D., Leybourne, S., & Newbold, P. (1997). Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13, 281–291. doi:

10.1016/S0169-2070(96)00719-4

- Hodson, T. O., & Hariharan, J. A. (2023). *dataretrieval (Python): a Python package for discovering and retrieving water data available from federal hydrologic web services*. U.S. Geological Survey software release. doi: 10.5066/P94I5TX3
- Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70. Retrieved from <https://www.jstor.org/stable/4615733>
- Jia, X., Zwart, J., Sadler, J., Appling, A., Oliver, S., Markstrom, S., ... Kumar, V. (2021). Physics-guided recurrent graph model for predicting flow and temperature in river networks. In *Proceedings of the 2021 siam international conference on data mining (sdm)* (pp. 612–620). SIAM. (Preprint: arXiv:2009.12575) doi: 10.1137/1.9781611976700.69
- Kavetski, D., Kuczera, G., & Franks, S. W. (2006a). Bayesian analysis of input uncertainty in hydrological modeling: 1. theory. *Water Resources Research*, 42, W03407. doi: 10.1029/2005WR004368
- Kavetski, D., Kuczera, G., & Franks, S. W. (2006b). Bayesian analysis of input uncertainty in hydrological modeling: 2. application. *Water Resources Research*, 42, W03408. doi: 10.1029/2005WR004376
- Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., ... Liu, T.-Y. (2017). LightGBM: a highly efficient gradient boosting decision tree. In *Advances in neural information processing systems 30 (neurips 2017)* (pp. 3146–3154). Retrieved from <https://proceedings.neurips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html>
- Kingma, D. P., & Ba, J. (2015). Adam: a method for stochastic optimization. In *International conference on learning representations (iclr)*.
- Künsch, H. R. (1989). The jackknife and the bootstrap for general stationary observations. *The Annals of Statistics*, 17, 1217–1241. doi: 10.1214/aos/1176347265
- Loshchilov, I., & Hutter, F. (2019). Decoupled weight decay regularization. In *International conference on learning representations (iclr)*.
- Luo, S., Yu, R., Chen, S., Fan, Y., Xie, Y., Li, Y., & Jia, X. (2025). Geo-aware models for stream temperature prediction across different spatial regions and scales. In *Proceedings of the 33rd acm international conference on advances in*

- geographic information systems (gispatial '25)* (pp. 124–136). ACM. (Preprint: arXiv:2510.09500) doi: 10.1145/3748636.3762716
- Martins, A. F. T., & Astudillo, R. F. (2016). From softmax to sparsemax: a sparse model of attention and multi-label classification. In *Proceedings of the 33rd international conference on machine learning (icml)* (pp. 1614–1623). Retrieved from <https://proceedings.mlr.press/v48/martins16.html>
- Mohseni, O., Stefan, H. G., & Erickson, T. R. (1998). A nonlinear regression model for weekly stream temperatures. *Water Resources Research*, *34*, 2685–2692. doi: 10.1029/98WR01877
- Nash, J. E., & Sutcliffe, J. V. (1970). River flow forecasting through conceptual models part I — a discussion of principles. *Journal of Hydrology*, *10*, 282–290. doi: 10.1016/0022-1694(70)90255-6
- Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., ... Chintala, S. (2019). PyTorch: an imperative style, high-performance deep learning library. In *Advances in neural information processing systems 32 (neurips 2019)* (pp. 8024–8035).
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ... Duchesnay, É. (2011). Scikit-learn: machine learning in Python. *Journal of Machine Learning Research*, *12*, 2825–2830.
- Piccolroaz, S., Calamita, E., Majone, B., Gallice, A., Siviglia, A., & Toffolon, M. (2016). Prediction of river water temperature: a comparison between a new family of hybrid models and statistical approaches. *Hydrological Processes*, *30*, 3901–3917. doi: 10.1002/hyp.10913
- Rahmani, F., Appling, A., Feng, D., Lawson, K., & Shen, C. (2023). Identifying structural priors in a hybrid differentiable model for stream water temperature modeling. *Water Resources Research*, *59*, e2023WR034420. doi: 10.1029/2023WR034420
- Rahmani, F., Lawson, K., Ouyang, W., Appling, A., Oliver, S., & Shen, C. (2021). Exploring the exceptional performance of a deep learning stream temperature model and the value of streamflow data. *Environmental Research Letters*, *16*, 024025. doi: 10.1088/1748-9326/abd501
- Rahmani, F., Shen, C., Oliver, S., Lawson, K., & Appling, A. (2021). Deep learning approaches for improving prediction of daily stream temperature in data-

- scarce, unmonitored, and dammed basins. *Hydrological Processes*, *35*, e14400. doi: 10.1002/hyp.14400
- Richardson, D. S. (2000). Skill and relative economic value of the ECMWF ensemble prediction system. *Quarterly Journal of the Royal Meteorological Society*, *126*, 649–667. doi: 10.1002/qj.49712656313
- Romano, Y., Patterson, E., & Candès, E. J. (2019). Conformalized quantile regression. In *Advances in neural information processing systems 32 (neurips 2019)* (pp. 3538–3548). Retrieved from <https://papers.nips.cc/paper/2019/hash/5103c3584b063c431bd1268e9b5e76fb-Abstract.html>
- Sadler, J. M., Appling, A. P., Read, J. S., Oliver, S. K., Jia, X., Zwart, J. A., & Kumar, V. (2022). Multi-task deep learning of daily streamflow and water temperature. *Water Resources Research*, *58*, e2021WR030138. doi: 10.1029/2021WR030138
- Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q., Hinton, G., & Dean, J. (2017). Outrageously large neural networks: the sparsely-gated mixture-of-experts layer. In *International conference on learning representations (iclr)*. Retrieved from <https://arxiv.org/abs/1701.06538>
- Thornton, M. M., Shrestha, R., Wei, Y., Thornton, P. E., & Kao, S.-C. (2022). *Daymet: daily surface weather data on a 1-km grid for North America, version 4 R1*. ORNL DAAC, Oak Ridge, Tennessee, USA. doi: 10.3334/ORNLDAAAC/2129
- Toffolon, M., & Piccolroaz, S. (2015). A hybrid model for river water temperature as a function of air temperature and discharge. *Environmental Research Letters*, *10*, 114011. doi: 10.1088/1748-9326/10/11/114011
- U.S. Geological Survey. (2024). *USGS water data for the nation: U.S. geological survey national water information system (NWIS) database*. Web services accessed via the dataretrieval Python client; parameter codes 00010, 00060, 00065 with statistic code 00003. (Accessed 2026; <https://waterdata.usgs.gov/nwis>) doi: 10.5066/F7P55KJN
- Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., ... SciPy 1.0 Contributors (2020). SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, *17*, 261–272. doi: 10.1038/s41592-019-0686-2

- Vovk, V., Gammerman, A., & Shafer, G. (2005). *Algorithmic learning in a random world*. New York: Springer. doi: 10.1007/b106715
- Weierbach, H., Lima, A. R., Willard, J. D., Hendrix, V. C., Christianson, D. S., Lubich, M., & Varadharajan, C. (2022). Stream temperature predictions for river basin management in the pacific northwest and mid-atlantic regions using machine learning. *Water*, *14*, 1032. doi: 10.3390/w14071032
- Wilks, D. S. (2011). *Statistical methods in the atmospheric sciences* (3rd ed., Vol. 100). Oxford: Academic Press.
- Zwart, J. A., Diaz, J., Hamshaw, S., Oliver, S., Ross, J. C., Sleckman, M., . . . White, E. (2023). Evaluating deep learning architecture and data assimilation for improving water temperature forecasts at unmonitored locations. *Frontiers in Water*, *5*, 1184992. doi: 10.3389/frwa.2023.1184992